THE U-BOAT WAR

## SINKING OF ROYAL OAK

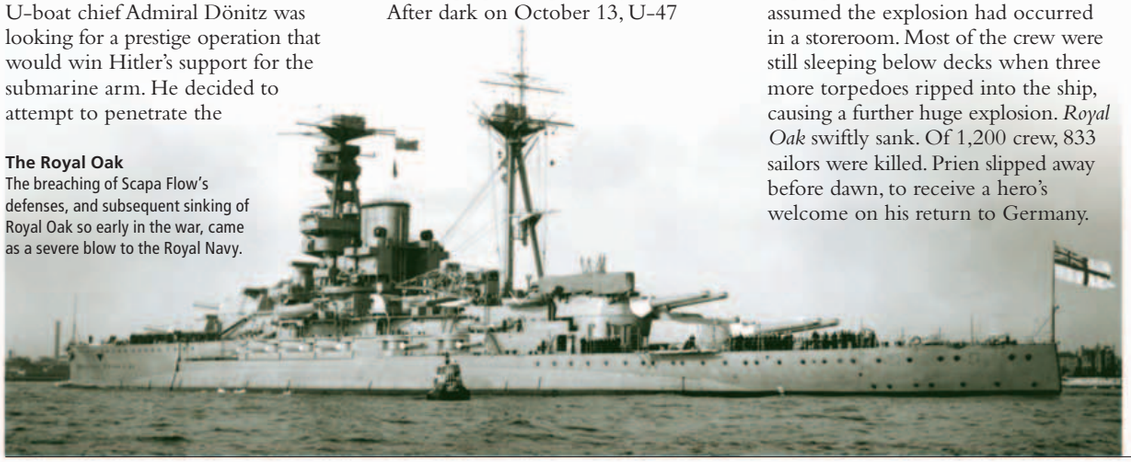


**Date** October 14, 1939
**Forces** British: 1 battleship; Germans: 1 submarine
**Losses** British: 1 battleship; Germans: none

**Location** Scapa Flow, Orkney Islands

At the start of World War II German U-boat chief Admiral Dönitz was looking for a prestige operation that would win Hitler's support for the submarine arm. He decided to attempt to penetrate the British deep-water anchorage at Scapa Flow. Although Scapa was well defended against submarine attack, German reconnaissance had shown narrow channels between the sunken blockships at the eastern end and Dönitz believed these were navigable by a U-boat on the surface. Lieutenant Günther Prien sailed U-47 out of Kiel on October 8 and proceeded across the North Sea, traveling on the surface by night and submerging during the daylight hours. After dark on October 13, U-47 squeezed between the blockships, carried forward by a strong current. Prien emerged into the harbor after midnight to find it largely deserted. Most of the British fleet had left for fear of air attack and ageing battleship *Royal Oak* had been left behind, moored next to a seaplane carrier.

With his first volley of torpedoes Prien hit *Royal Oak* near the bows. The Royal Navy believed Scapa Flow impregnable to U-boats, so it was assumed the explosion had occurred in a storeroom. Most of the crew were still sleeping below decks when three more torpedoes ripped into the ship, causing a further huge explosion. *Royal Oak* swiftly sank. Of 1,200 crew, 833 sailors were killed. Prien slipped away before dawn, to receive a hero's welcome on his return to Germany.



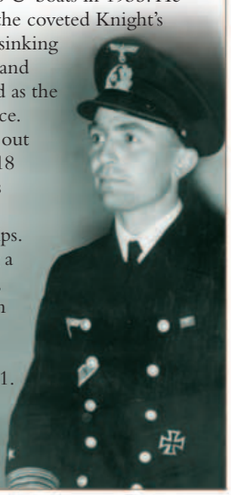
**The Royal Oak**
The breaching of Scapa Flow's defenses, and subsequent sinking of Royal Oak so early in the war, came as a severe blow to the Royal Navy.

1908–1941

## GÜNTHER PRIEN

**COMMANDER OF GERMAN SUBMARINE U-47**

U-boat commander Günther Prien originally trained as a merchant seamen. He joined the Reichsmarine and the Nazi Party in the early 1930s, transferring to U-boats in 1935. He was awarded the coveted Knight's Cross for the sinking of *Royal Oak* and was celebrated as the first U-boat ace. Prien carried out 12 patrols in 18 months and is credited with sinking 30 ships. After refusing a shore posting, he went down with U-47 in the Atlantic in March 1941.



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## CONVOY SC-7



**Date** October 16–19, 1940
**Forces** British: 5 escort ships, 35 merchant ships; Germans: 7 U-boats
**Losses** British: 20 merchant ships; Germans: none

**Location** Western Approaches, Atlantic

SC-7 was a convoy of slow merchant ships that set out from Nova Scotia for Britain on October 5, 1940, with a single sloop as escort. Although another two sloops and two corvettes met the convoy as it entered the Western Approaches, none of them had any idea how to cope with multiple attacks by submarines operating on the surface at night. The convoy itself was undisciplined, with too many ships out of formation. Four merchant ships were sunk by U-boats on October 16 and 17, before the main attack commenced on the night of October 18-19. This involved five U-boats, including Otto Kretschmer's U-99, Joachim Schepke's U-100, and Engelbert Endrass's U-46. They sank 16 ships in six hours, while the escorts were reduced to picking up survivors.

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## CONVOY HX-79



**Date** October 19–20, 1940
**Forces** British: 11 escort ships, 49 merchant ships; Germans: 5 U-boats
**Losses** British: 12 merchant ships; Germans: none

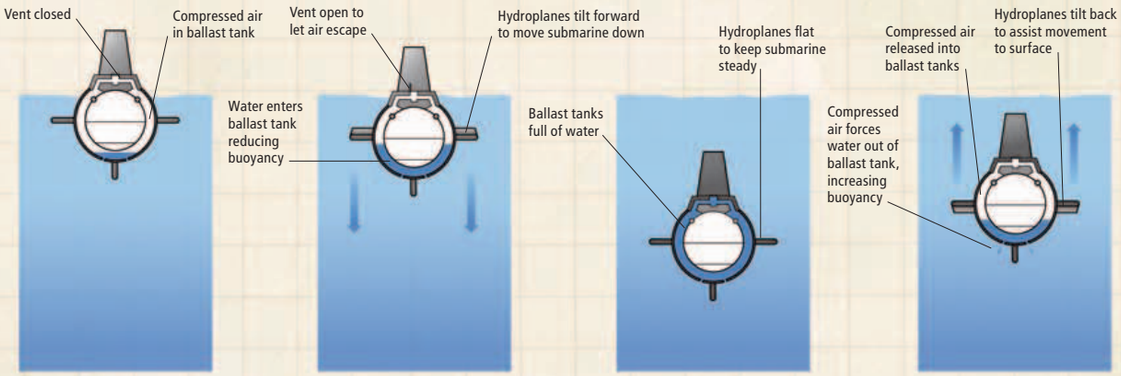
**Location** Western Approaches, Atlantic

Shocked by the savaging of Convoy SC-7 on October 19, 1940, the British Admiralty rushed out 11 warships to escort another convoy, HX-79, then approaching Britain. The escorts included two destroyers and four corvettes. Meanwhile, HX-79 was being shadowed by Gunther Prien's U-47. Four U-boats that had participated in the attack on SC-7, including Joachim Schepke's U-100 and Engelbert Endrass's U-46, joined Prien. That night the U-boats slipped through the escort screen on the surface and attacked the merchant vessels. Ten ships sank in the darkness, four of them claimed by Prien. Two more ships sank the next day, bringing total British losses over five days in the Atlantic to 32 ships totaling 150,000 tons. No U-boats were damaged.

WEAPONS AND TECHNOLOGY

## HOW SUBMARINES WORK

Depending on its density, an object will either float or sink in water. When a submarine is about to dive, vents at the top of the hull are opened, air escapes, and the reduced pressure allows water in through vents in the bottom. The density of the submarine increases and it sinks. With the aid of hydroplanes, the engine powers the submarine forward and down until the ballast tanks are completely full and a state of neutral buoyancy has been achieved. At this point the upper vents are closed. To reverse the process, compressed air is forced into the ballast tanks, forcing water out of the lower vents and increasing buoyancy, while the engine propels the submarine back to the surface.




**Partially submerged**
Ballast tanks filled with compressed air allow the submarine to float just below the surface of the water.

**Submerging**
As air is released the ballast tanks fill with water and the submarine starts to submerge.

**Fully submerged**
The submarine continues to sink until the ballast tanks are completely filled with water.

**Surfacing**
Compressed air released into the ballast tanks forces water out and the submarine surfaces.